Lab 5 – Open Daylight SDN Controller

Student Full Name_______ Greemesh Basnet_______ _________/10 Marks

Objective

To use a SDN Controller to gather information about a network and configure network settings.

Background

- Use the material posted from weeks 6 and 7 to identify the architecture of a Software Defined Network.
- Mininet is a network emulator which creates a network of virtual hosts, switches, controllers, and links. Mininet hosts run standard Linux network software, and its switches support OpenFlow for highly flexible custom routing and Software-Defined Networking. Mininet provides an extensible Python API for network creation and experimentations. Mininet enables:
  - Rapid prototyping of software-defined networks
  - Complex topology testing without the need to wire up a physical network
  - Multiple concurrent developers to work independently on the same topology
- Carbon is the sixth release of OpenDaylight (ODL), the leading open source platform for programmable, Software-Defined Networks. Fluorine is the ninth release of OpenDaylight. With this version, the project has moved to a system of managed releases, with streamlined packaging to accelerate the development and deployment of OpenDaylight-based solutions.
- In this lab, you first install Mininet on a VM and then OpenDaylight on another VM to prepare an environment to do some experiments on SDN.

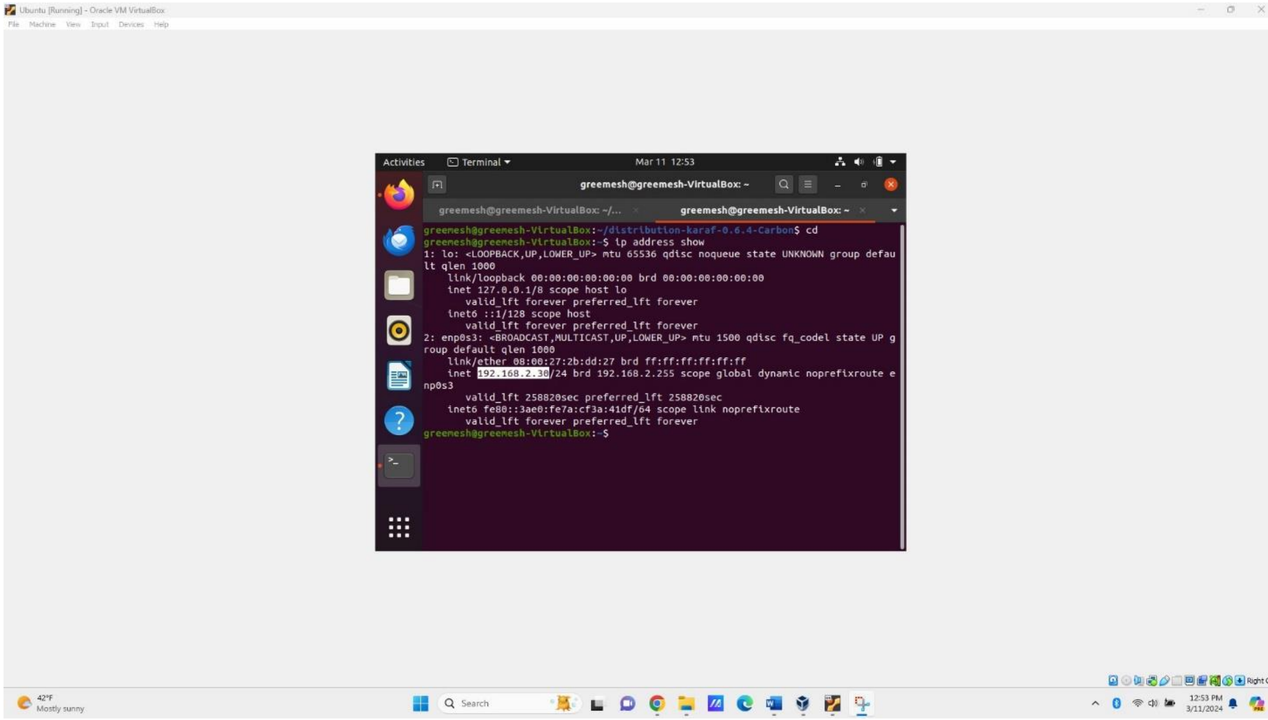
Instructions

1. Use a different font and color to write your answers.
2. Do not erase anything from this document.
3. Insert your name on each screenshot. Resize to make them easily readable, and include the date and time.

Mininet installation:

1- Visit the Mininet Webpage: https://github.com/mininet/mininet/releases/
2- From the section Assets download: mininet-2.3.0-210211-ubuntu-16.04.7-server-amd64-ovf.zip
3- Create a new VM with this image:
    a. Use the Network setting 'Bridge' on your network card.

b. What is IP address in this VM? ____198.168.2.30___ **Take the screenshot showing the IP configuration of your VM. [One mark]**



OpenDayLight installation:

1- Download Ubuntu Desktop image (22.04) from: https://ubuntu.com/download/desktop#download. Select the 22.04 LTS distribution and create one Ubuntu VM.

2- Go to the OpenDaylight website:
https://docs.opendaylight.org/en/stable-argon/getting-started-guide/installing_opendaylight.html
Follow the link to the software download page and then from the Archived Releases select '*Carbon and earlier*'. Select '*0.6.4-Carbon*' and then Download this OpenDaylight distribution:
distribution-karaf-0.6.4-Carbon.zip

Follow the instructions on this website. Share a folder with VM and copy this file into home directory.

3- Unzip the file.

4- Run OpenDaylight controller by going to the unzipped file directory and run:
./bin/karaf
(In case you need to install Java then run: sudo apt install openjdk-8-jdk)

(You can set JAVA_HOME here: export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64)

Useful Links:

https://vitux.com/how-to-setup-java_home-path-in-ubuntu/

https://www.brianlinkletter.com/2016/02/using-the-opendaylight-sdn-controller-with-the-mininet-network-emulator/

5- Install these features:

odl-restconf

odl-l2switch-switch

odl-mdsal-apidocs

odl-dlux-core

odl-dluxapps-nodes

odl-dluxapps-topology

odl-dluxapps-yangui

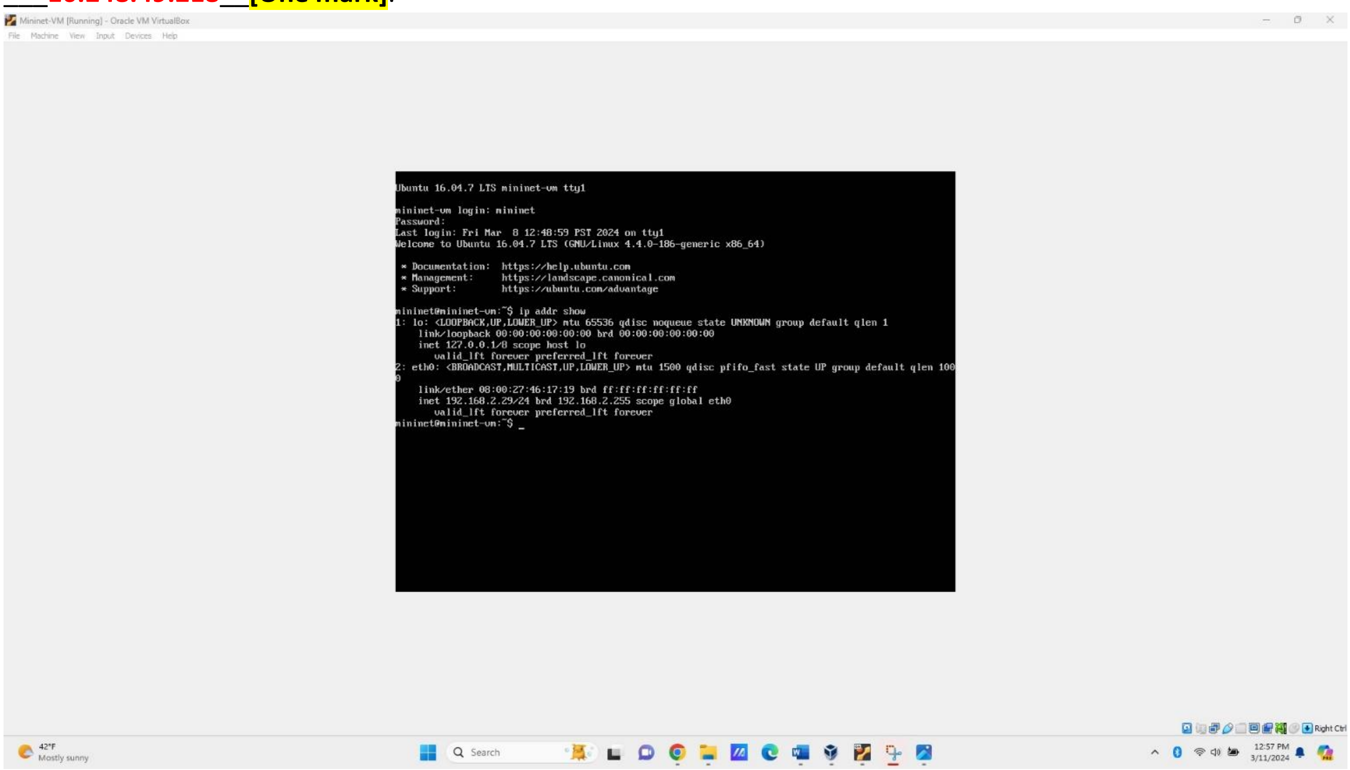
odl-dluxapps-yangvisualizer

odl-dluxapps-yangman

Be sure you enter them as separate commands using the **feature:install** command prior to each of them, here you can see installing some of them. See the below image:

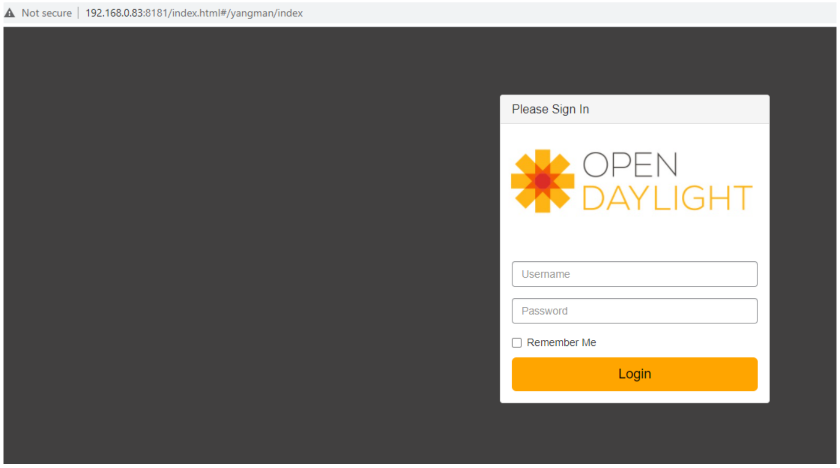
```
opendaylight-user@root>feature:install odl-dlux-core
opendaylight-user@root>feature:install odl-dluxapps-nodes
opendaylight-user@root>feature:install odl-dluxapps-yangui
opendaylight-user@root>feature:install odl-dluxapps-yangvisualizer
opendaylight-user@root>feature:install odl-dluxapps-yangman
opendaylight-user@root>
```

6- Network setting should be Bridge on your network card, same as Mininet.
7- What is the <IP address> of this VM? **Take the screenshot showing the IP configuration of your VM** ___10.248.49.218__**[One mark]**.

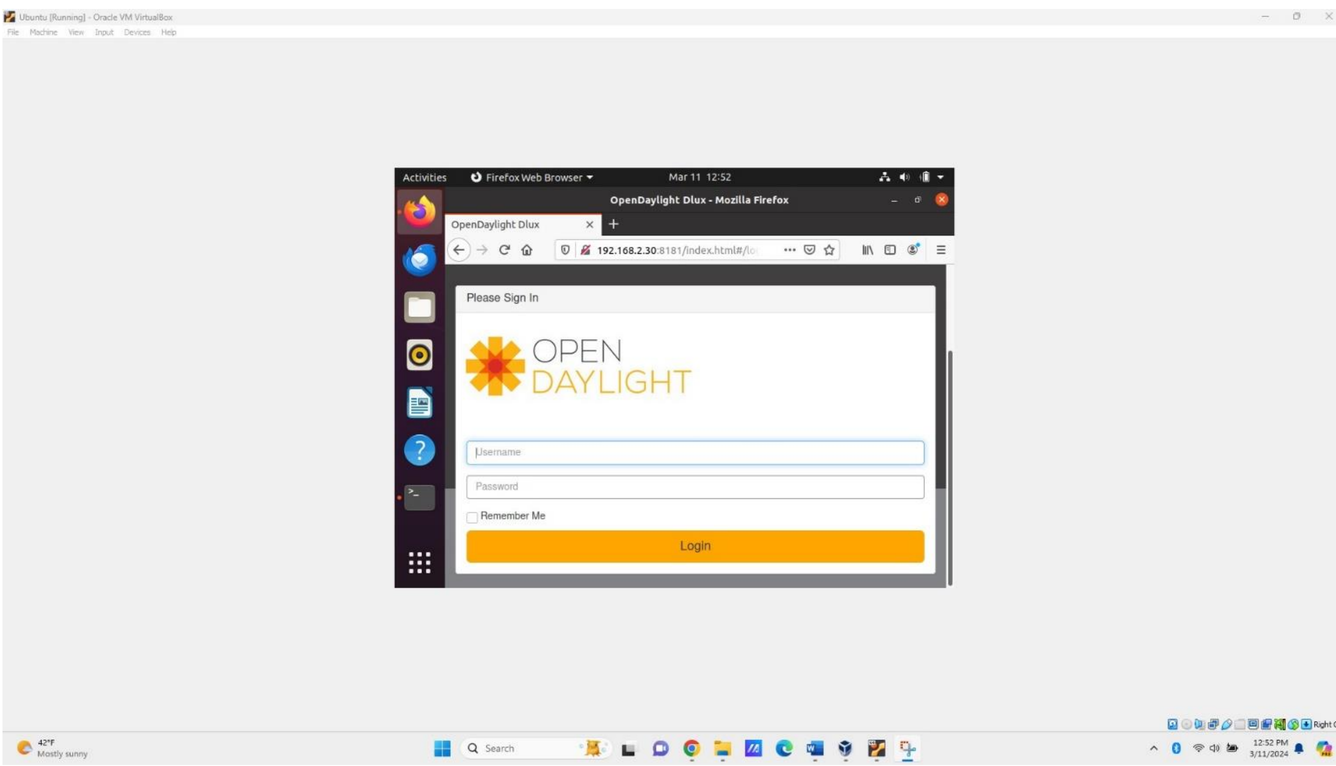


8- Use it for the next step.
9- Login to Controller web interface: <ip address>: 8181/index.html
(as an example: http://192.168.0.83:8181/index.html)

Default user/pass is “**admin/admin**”.



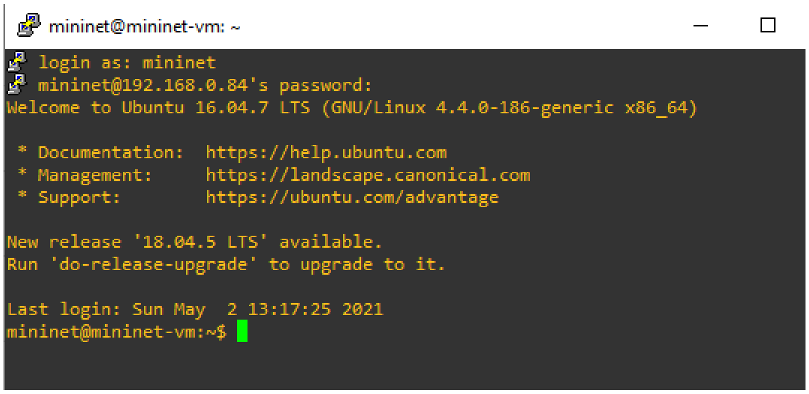
Replace this image with the screenshot of your own Open Daylight screen. **Be sure that the <u>IP address</u> on the top bar is visible, as well as <u>your name</u> and the <u>date and time</u>. [Two marks]**



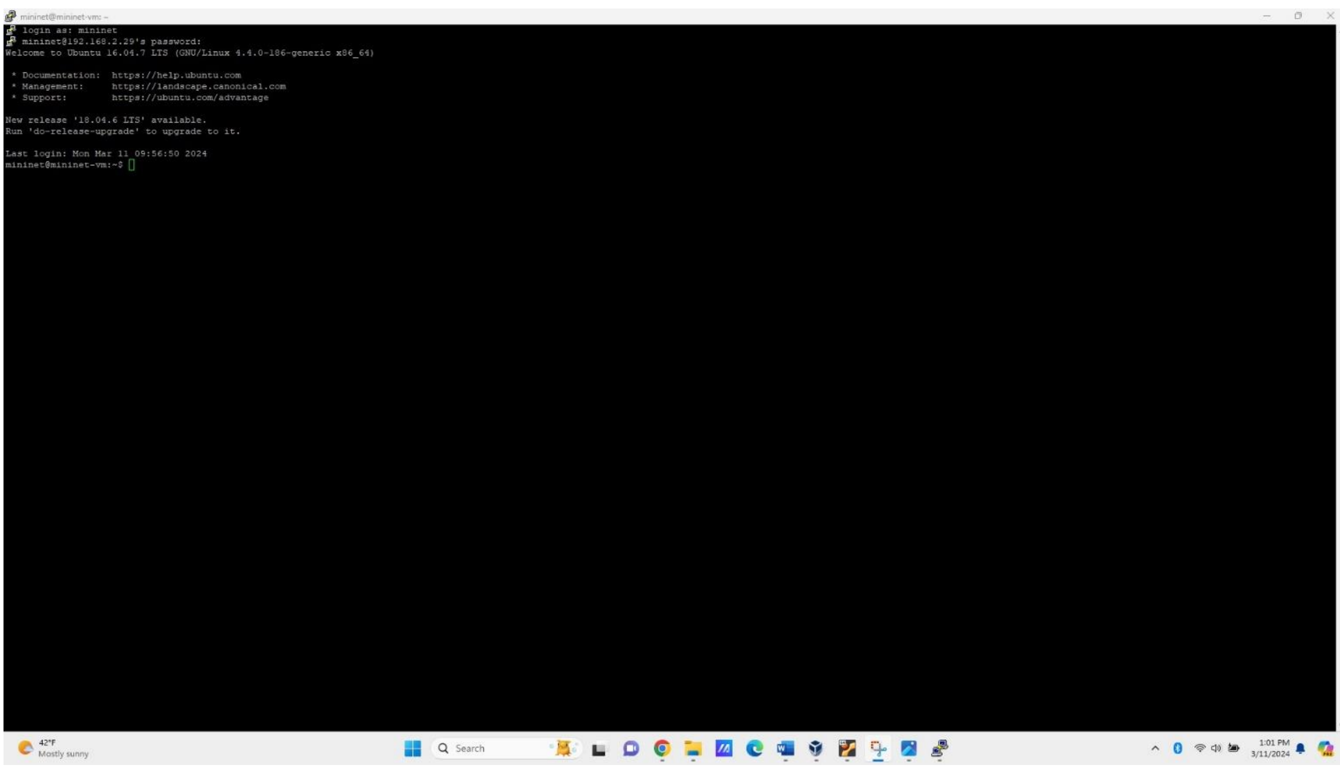
<u>Connect Mininet and Controller:</u>

1- Start Mininet VM (default user/pass is “**mininet/mininet**”)

2- Open Putty and do SSH connection to Mininet

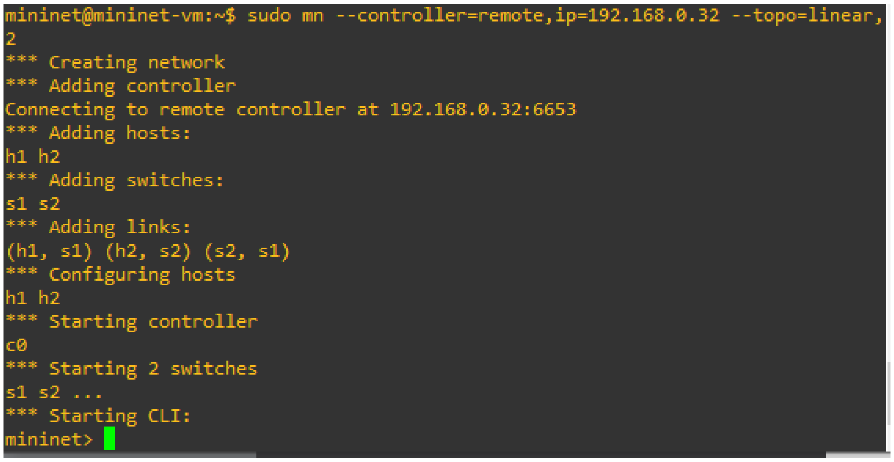


Replace this image with the screenshot of your own Mininet screen. **Be sure that the top bar is visible, as well as your <u>name</u> and the <u>date and time</u>. [Two marks]**

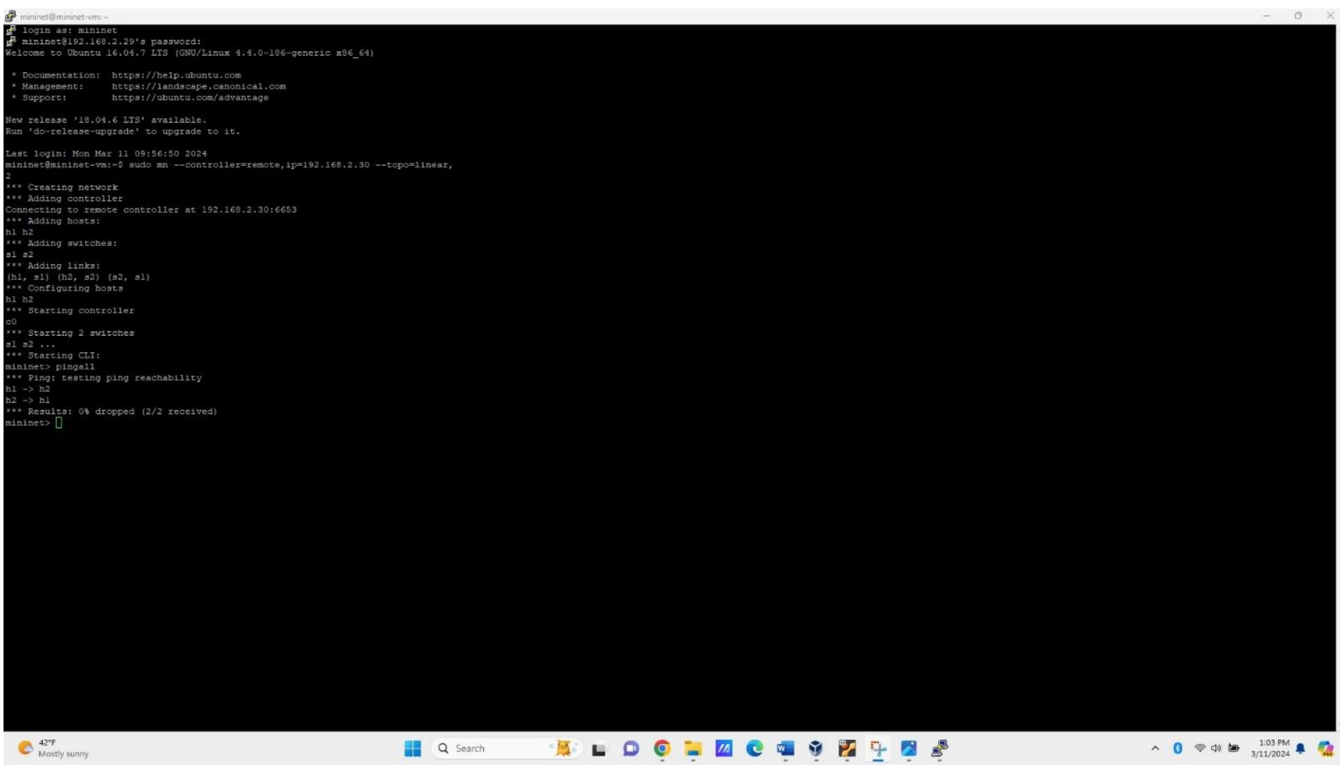


Run this command to configure an example network.

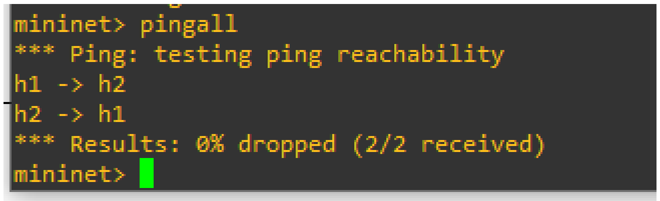
sudo mn --controller=remote,ip=<controller IP address> --topo=linear,2



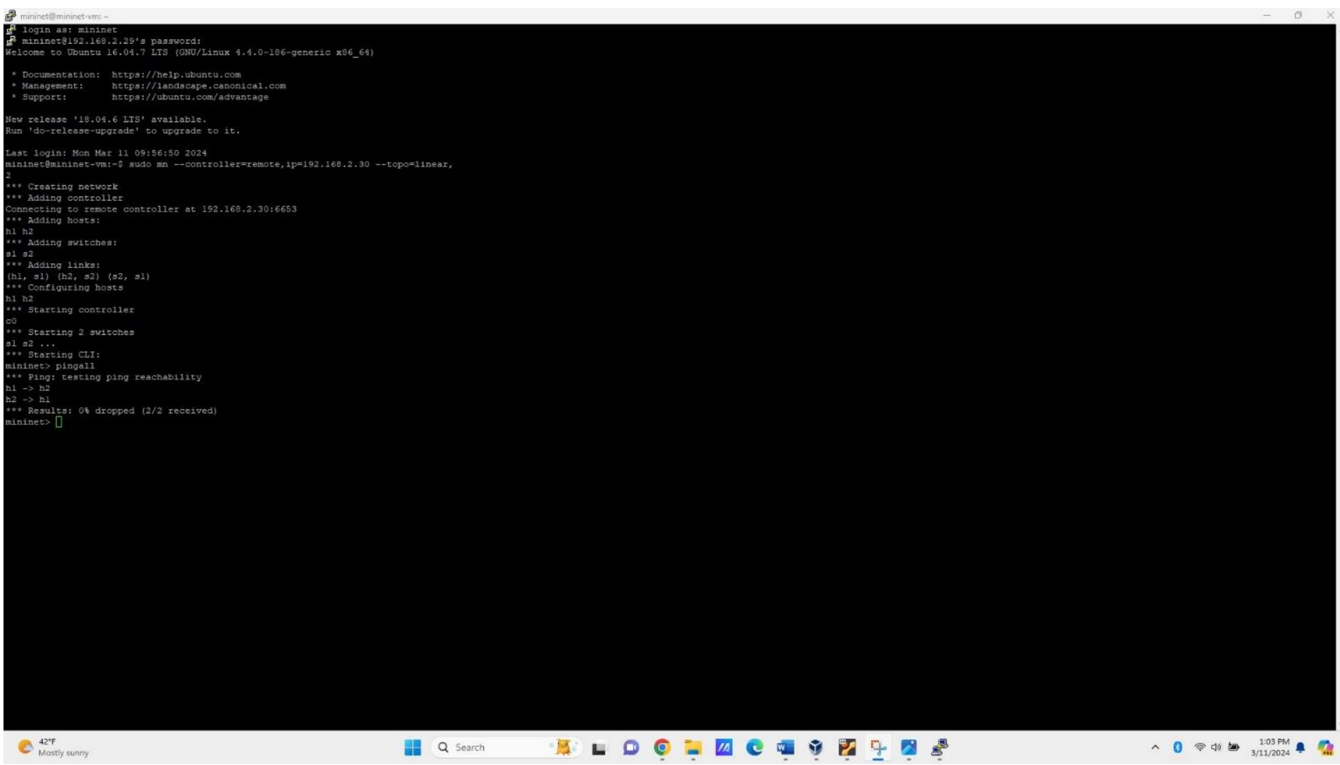
Replace this image with the screenshot of your own Mininet screen. **Be sure that the top bar is visible, as well as your <u>name</u> and the <u>date and time</u>.** **[One mark]**



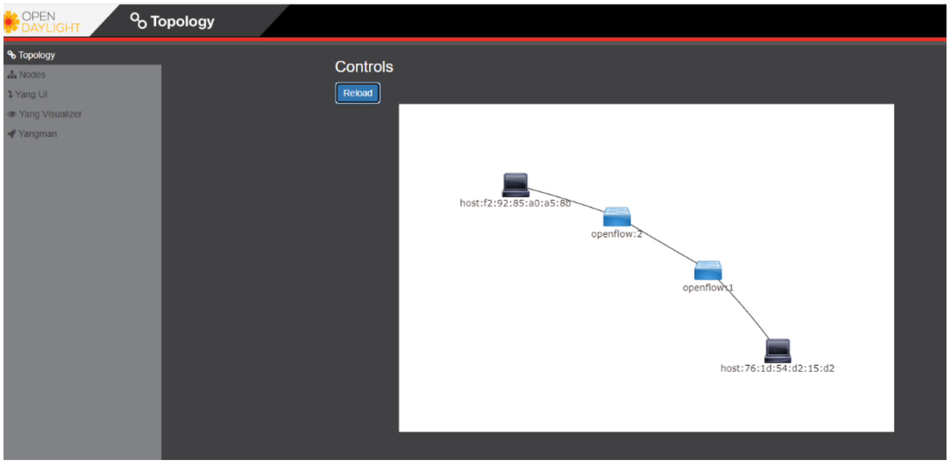
Run: pingall



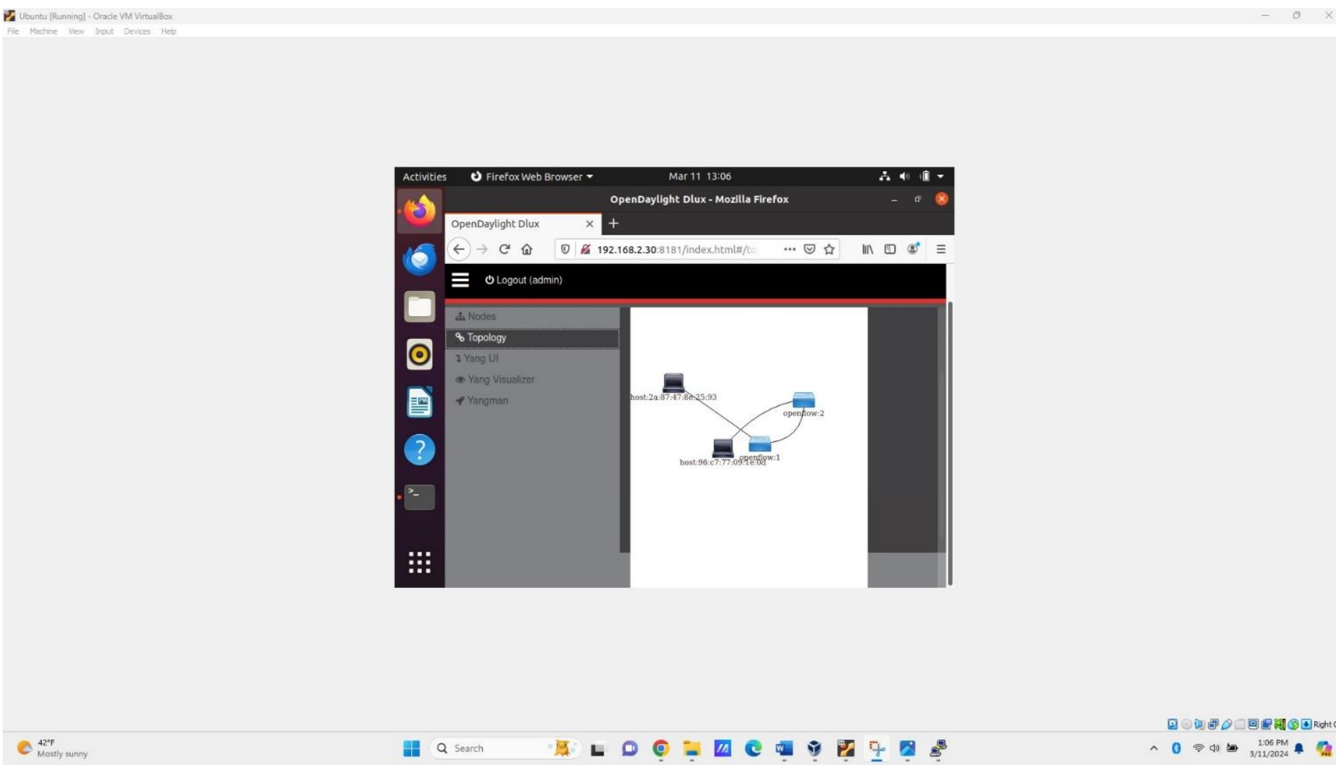
Replace this image with the screenshot of your own Mininet screen. **Be sure that the top bar is visible, as well as your <u>name</u> and the <u>date and time</u>.** **[One mark]**



And now in the Open Daylight controller web interface and topology page, you should see the example network:



Replace this image with the screenshot of your own Open Daylight screen. **Be sure that the top bar is visible, as well as your <u>name</u> and the <u>date and time</u>. [Two marks]**



To submit:

1. Be sure that you wrote your full name at the beginning of this document. (10% deducted if you do not write your name)

2. Rename this report file as **LastNameFirstName_CSN305Lab5** and save it as PDF. Upload this document including all your answers and screenshots. (10% deducted if this file is missed in your submission or wrong name is used)